Bio 1320 — *Lulu's* Exam 2 Study Guide

Chapters 6 · 7 · 8 · Cancer | Essentials of the Living World — Johnson & Bergh

Exam 2 — March 30, 2026

How to Use This Guide

- **Gold boxes** = Key facts — highest priority for the exam
- **Green boxes** = Definitions — exact vocabulary
- **Red boxes** = Watch out — common mistakes and tricky points
- **Dark boxes** = Equations — know these cold
- Mnemonics are collected at the very end — use them or skip them

6	7	8A	8B
Photosynthesis	Cell Respiration	Mitosis	Cancer

6 Chapter 6: Photosynthesis

Light Energy · Chloroplasts · Light Reactions · Calvin Cycle

The Master Equation

PHOTOSYNTHESIS OVERALL EQUATION

$$6CO_2 + 6H_2O + \text{Light Energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

Carbon Dioxide + Water + Light → Glucose + Oxygen

KEY FACTS

- Photosynthesis is the process by which solar energy is trapped and stored as chemical energy in the bonds of glucose
- **Endergonic reaction** — requires energy input (light) to run
- Takes place in **chloroplasts** (found in plant cells and algae)
- Autotrophs (producers) carry out photosynthesis — essential to sustaining all life on Earth
- The overall reaction is the reverse of cellular respiration

Plant Anatomy & Chloroplast Structure

KNOW WHERE EACH REACTION OCCURS — THIS IS A COMMON EXAM QUESTION

Structure	Location	Function
Mesophyll cells	Middle layer of leaf	Main site of photosynthesis; each cell contains several chloroplasts
Stomata	Pores on leaf surface	Gas exchange — CO_2 enters, O_2 exits; water vapor released
Thylakoid membrane	Stacked disc membranes inside chloroplast	**Light-dependent reactions** occur here

Granum (pl: grana)	Stack of thylakoids	Houses photosystems and the electron transport chain
Stroma	Fluid surrounding thylakoids	**Calvin cycle (light-independent reactions)** occur here

Light & Photosynthetic Pigments

KEY LIGHT FACTS

- The sun emits energy in waves — **shorter wavelengths have higher energy** than longer wavelengths
- Different pigments specialize in absorbing different wavelengths
- **Chlorophyll a** is the primary photosynthetic pigment — absorbs mostly **blue and red light**
- Green light is reflected (not absorbed) — this is why plants appear green
- A **photosystem** is a large protein structure in the thylakoid membrane consisting of antenna pigments and a reaction center containing chlorophyll

Stage 1: Light-Dependent Reactions

DEFINITION

Light-dependent reactions = the first stage of photosynthesis. Uses light energy to split water and produce ATP and NADPH. Occurs in the **thylakoid membrane**.

WHAT HAPPENS — TWO PHOTOSYSTEMS WORKING IN SEQUENCE

Photosynthesis involves 2 processes: (1) light-dependent — splitting of H_2O to form O_2 and H; (2) light-independent — combining H and CO_2 to form sugar.

Step	Photosystem II (PSII)	Electron Transport Chain	Photosystem I (PSI)
What happens	Light hits chlorophyll → water is split → O_2 released → electrons energized	Energized electrons pass through carrier molecules → protons pumped → ATP made via chemiosmosis	Electrons re-energized by light → $NADP^+$ reduced → forms NADPH

Products	O_2 (released) + energized electrons	ATP (via ATP synthase)	NADPH

NOTE — PHOTOSYSTEM NUMBERING

Photosystem II comes **before** Photosystem I in the actual sequence. They were named in the order they were discovered, not the order they work.

KEY TERMS — LIGHT REACTIONS

- **Chemiosmosis** = protons flow through ATP synthase to generate ATP
- **ATP synthase** = the protein channel protons flow through; makes ATP as they pass
- **NADPH** = energy carrier that carries hydrogen atoms to the Calvin cycle
- **Photolysis** = splitting of water by light energy in PSII

Stage 2: The Calvin Cycle (Light-Independent Reactions)

DEFINITION

Calvin Cycle = the second stage of photosynthesis. Uses ATP and NADPH from the light reactions to convert CO_2 into glucose. Occurs in the **stroma**.

REACTANTS & PRODUCTS

REACTANTS (in):

- CO_2 (from air through stomata)
- ATP (from light reactions)
- NADPH (from light reactions)

PRODUCTS (out):

- G3P → Glucose ($C_6H_{12}O_6$)
- ADP + P_i (returned to light reactions)
- $NADP^+$ (returned to light reactions)

THE 3 STEPS OF THE CALVIN CYCLE

1. **Carbon Fixation:** CO_2 is attached to RuBP (5-carbon) by the enzyme **RuBisCO** → makes a 6-carbon compound that immediately splits into two 3-carbon molecules (3-PGA)
2. **Reduction:** ATP and NADPH are used to convert 3-PGA into **G3P** (glyceraldehyde-3-phosphate) — the actual sugar product
3. **Regeneration of RuBP:** Most G3P is used to regenerate RuBP (using ATP) so the cycle can continue; some G3P exits to make glucose

ROLE OF ATP AND NADPH IN THE CALVIN CYCLE

ATP provides the energy and NADPH provides the hydrogen atoms needed to reduce CO_2 into G3P (sugar). Both come from the light reactions and are recycled back as ADP and $NADP^+$ when the Calvin cycle runs.

7 Chapter 7: Cellular Respiration

How Cells Release Energy · Glycolysis · Krebs Cycle · ETC · Fermentation

The Master Equation

CELLULAR RESPIRATION OVERALL EQUATION

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 36\ ATP$$

Glucose + Oxygen → Carbon Dioxide + Water + Energy (ATP)

KEY FACTS

- Molecules in food (such as glucose) are used to produce ATP — the energy carrier used to power cell activities
- Cells cannot survive without ATP
- **Exergonic reaction** — releases energy stored in glucose as ATP
- Aerobic cellular respiration is used by all plants and animals, as well as many microbes
- Occurs in the **cytoplasm + mitochondria**
- Do not confuse with breathing — cellular respiration is a chemical process inside cells

Photosynthesis vs. Cellular Respiration

Photosynthesis	Cellular Respiration
Uses CO_2 + H_2O + Light	Uses Glucose + O_2
Produces Glucose + O_2	Produces CO_2 + H_2O + ATP
Endergonic (requires energy)	Exergonic (releases energy)
Occurs in chloroplasts	Occurs in cytoplasm + mitochondria
Only in plants and algae	In ALL living cells (plants too)

WHERE Each Step Happens — Know the Locations

Stage	Location	Requires O_2?	ATP Made
Glycolysis	**Cytoplasm** (NOT in mitochondria)	No	**2 ATP**
Pyruvate → Acetyl-CoA (Transition)	**Mitochondrial matrix**	Yes	0 ATP (makes NADH)
Krebs Cycle	**Mitochondrial matrix**	Yes	**2 ATP**
Electron Transport Chain (ETC)	**Inner mitochondrial membrane**	Yes — O_2 is the final electron acceptor	**34 ATP**
Fermentation	**Cytoplasm**	No (anaerobic)	0 additional (only glycolysis ATP = 2)

TOTAL ATP FROM 1 GLUCOSE = 36 ATP

Glycolysis: **2** + Krebs: **2** + ETC: **34** = **36 ATP total**

Fermentation only = 2 ATP (just from glycolysis — no Krebs or ETC)

Stage 1: Glycolysis — Splitting Glucose

DEFINITION

Glycolysis = "glucose splitting" — the breakdown of 1 glucose molecule into 2 pyruvate molecules. Occurs in the **cytoplasm**. Does not require oxygen. Happens in all cells (aerobic and anaerobic). "Glycolysis means splitting of glucose."

GLYCOLYSIS SUMMARY

REQUIRES:

- 1 glucose
- 2 NAD^+
- 2 ADP

YIELDS:

- 2 pyruvate
- 2 NADH (electron carriers)
- **2 ATP** (net gain)

Stage 2: Pyruvate → Acetyl-CoA (Transition Step)

WHAT HAPPENS

- Each pyruvate (3-carbon) loses a CO_2 → becomes a 2-carbon acetyl group
- The acetyl group attaches to CoA → forms **Acetyl-CoA**
- NAD^+ is reduced to **NADH**
- CO_2 is released (part of the CO_2 you exhale)
- Occurs twice — once per pyruvate (2 pyruvates from glycolysis)
- Location: **mitochondrial matrix**

Stage 3: Krebs Cycle

DEFINITION

Krebs Cycle (Citric Acid Cycle) = a series of reactions in the **mitochondrial matrix** that breaks down Acetyl-CoA, releasing CO_2 and producing electron carriers (NADH, $FADH_2$) and a small amount of ATP.

KREBS CYCLE — PER 2 ACETYL-COA (BOTH TURNS OF THE CYCLE)

REACTANTS:

- 2 Acetyl-CoA (2-carbon each)
- Oxaloacetate (recycled — the "starter")

PRODUCTS:

- 4 CO_2 (released as gas)
- 6 NADH (electron carriers)
- 2 $FADH_2$ (electron carriers)
- **2 ATP**

CO_2 IS PRODUCED IN TWO PLACES

CO_2 is released during: (1) the **Transition step** (pyruvate → Acetyl-CoA) AND (2) the **Krebs cycle**. This CO_2 eventually diffuses out and is exhaled.

Stage 4: Electron Transport Chain (ETC)

DEFINITION

Electron Transport Chain = a series of protein complexes in the **inner mitochondrial membrane**. NADH and $FADH_2$ donate electrons, which release energy used to pump protons (H^+) across the membrane, driving ATP synthesis via **chemiosmosis**.

1. NADH and $FADH_2$ donate their electrons to the ETC
2. Electrons pass through protein complexes, releasing energy at each step
3. Energy pumps H^+ ions into the intermembrane space (building up concentration)
4. H^+ ions flow back through **ATP synthase** → produces **34 ATP**
5. At the end of the chain, electrons + H^+ combine with $\mathbf{O_2}$ → $\mathbf{H_2O}$ (water)

WHAT HAPPENS WITHOUT O_2?

Without O_2, electrons cannot be accepted at the end of the ETC → the chain stops → NADH cannot be recycled → fermentation occurs instead.

Fermentation — When O_2 Is Not Available

DEFINITION

Fermentation = an anaerobic process that regenerates NAD^+ from NADH so glycolysis can continue. Generates ATP only in glycolysis (2 ATP total). Occurs in the **cytoplasm**.

Type	What Happens	Products	Example
Alcoholic fermentation	NADH reduces pyruvate to ethanol; NAD^+ is re-created	Ethanol + CO_2	Yeast making beer/bread
Lactic acid fermentation	NADH reduces pyruvate to lactic acid; NAD^+ is re-created	Lactic acid	Muscle cells during intense exercise

OTHER FOOD MOLECULES

Proteins and fats can also enter the energy-extracting pathways — they enter at different points (fats enter as acetyl-CoA; proteins enter as various intermediates).

8A Chapter 8: Mitosis & Cell Division

Cell Cycle · Binary Fission · Chromosomes · PMAT · Cytokinesis

Overview — Cell Division

KEY CONCEPTS

- Cell division produces a continuous supply of replacement cells
- Each daughter cell receives a complete set of hereditary information from the parent cell and about half its cytoplasm
- **Asexual reproduction** — offspring formed from a single parent; offspring are genetically identical clones
- **Sexual reproduction** — offspring produced by fusion of gametes (sperm and egg) from two adults; requires meiosis
- **Mitosis** adds new cells; **apoptosis** (programmed cell death) removes them — allowing tissues to renew themselves

Chromosome Vocabulary — Know the Differences

MUST-KNOW TERMS

- **Chromosome** = a single, condensed DNA molecule + proteins (histones). Humans have 46 chromosomes (23 pairs).
- **Chromatin** = the loose, uncondensed form of DNA + histone proteins (what chromosomes look like during interphase)
- **Nucleosome** = DNA wrapped around 8 histone proteins — the basic unit of chromatin packaging
- **Sister chromatids** = the two identical copies of a chromosome joined at the centromere after DNA replication
- **Homologous chromosomes** = a pair of chromosomes — one from mom, one from dad — that carry genes for the same traits but may have different versions (alleles). NOT identical copies.
- **Centromere** = the region where sister chromatids are joined; also the attachment site for spindle microtubules
- **Telomeres** = regions at the ends of chromosomes essential for chromosome stability
- **Diploid (2n)** = having 2 sets of chromosomes. Human body cells = 2n = 46.
- **Haploid (n)** = having 1 set of chromosomes. Human gametes (sperm/egg) = n = 23.
- **Alleles** = enduring mutations inherited generation after generation; different versions of a gene (e.g., black, brown, or blond hair)

SISTER CHROMATIDS VS. HOMOLOGOUS CHROMOSOMES — COMMON EXAM CONFUSION

Sister Chromatids	Homologous Chromosomes
IDENTICAL copies of the same chromosome	Same chromosome type but from different parents — may have different alleles
Joined at the centromere	NOT joined — separate chromosomes
Created by DNA replication (S phase)	Inherited — one from mom, one from dad
Separated during ANAPHASE of mitosis	Separated during meiosis I (not mitosis)

HUMAN CHROMOSOMES

- A typical human cell has **23 pairs of chromosomes, for a total of 46**
- 22 pairs = autosomes
- 1 pair = sex chromosomes (different in males and females)

Binary Fission — Prokaryote Division

DEFINITION

Binary fission = how prokaryotes (bacteria) reproduce. An asexual process that replicates DNA and distributes it to two daughter cells. Much simpler than mitosis — prokaryotes have one circular chromosome and no nucleus.

BINARY FISSION STEPS

1. The circular chromosome is replicated; DNA attaches to the cell membrane
2. The cell membrane grows between the two DNA molecules
3. The cell elongates
4. The cell divides in half, forming two identical daughter cells

The Cell Cycle

CELL CYCLE VS. MITOSIS — THEY ARE NOT THE SAME

Cell cycle = the entire life of a cell (interphase + mitosis + cytokinesis)
Mitosis = just the nuclear division phase of the cell cycle (PMAT)

Phase	Stage	What Happens
INTERPHASE	**G_1 (Gap 1)**	Cell grows; makes proteins and organelles; normal cell activities. Protein synthesis occurs.
	S (Synthesis)	DNA is replicated — every chromosome is copied; cell now has double the DNA
	G_2 (Gap 2)	Cell continues to grow; produces proteins necessary for mitosis; DNA begins to wind more tightly around histones

MITOSIS	PMAT (4 phases)	Nucleus divides; chromosomes are separated into 2 identical sets
CYTOKINESIS	After telophase	Cytoplasm divides; 2 daughter cells are formed

G_0 PHASE

G_0 = the cell continues its function but does not replicate its DNA or divide. Most cells in your body are currently in G_0.

CHROMOSOME NUMBER IN G_2

If an organism has 36 chromosomes in G_2, the diploid number is still 36. In G_2, DNA has been replicated (72 chromatids total) but the chromosome NUMBER is still 36 — each chromosome has 2 sister chromatids joined at the centromere. Count centromeres = count chromosomes.

The 4 Phases of Mitosis — PMAT

Phase	What You See	Key Events
PROPHASE	Chromosomes condense and become visible; nuclear envelope breaks down	Duplicated chromosomes condense · Spindle microtubules form from centrioles · Nuclear envelope breaks down · Nucleolus disappears · Chromosomes captured by spindle microtubules (at kinetochores)
METAPHASE	Chromosomes line up in the middle of the cell	Chromosomes align along the cell's equator (metaphase plate) · Spindle fibers attach to centromeres · Best phase to count chromosomes
ANAPHASE	Sister chromatids are pulled apart to opposite poles	Centromeres split · Sister chromatids separate and become independent daughter chromosomes · Motor proteins in kinetochores pull chromatids apart · Cell elongates

TELOPHASE + Cytokinesis	Two new nuclei form; cell splits in two	Nuclear envelopes re-form around each set of chromosomes · Chromosomes uncoil back to chromatin · Spindle fibers disintegrate · Nucleoli reappear · Cytokinesis begins

Cytokinesis — Splitting the Cytoplasm

Feature	Animal Cells	Plant Cells
Mechanism	Cleavage furrow — a ring of microfilaments contracts and pinches the cell in two	Cell plate forms in the middle and grows outward to divide the cell
Why different?	Animal cells are flexible — can be pinched	Plant cells have a rigid cell wall — cannot be pinched; must build a new wall

CELL CYCLE CHECKPOINTS

Chemical checkpoints regulate the cell cycle. These checkpoints ensure DNA is properly replicated and the cell is ready before proceeding to the next phase.

8B

Chapter 8: Cancer

What It Is · How It Makes You Sick · Staging · Treatments

What Is Cancer?

DEFINITION

Cancer = a disease of uncontrolled cell growth. Caused by genetic mutations that regulate cell division. Close to 40% of Americans will be diagnosed with cancer.

KEY FACTS

- Cancer is caused by **mutations in genes** that regulate cell division
- Most cancers require **multiple mutations** to develop
- Cancer cells do not respond to normal cell cycle checkpoints
- Cancer cells can be immortal — they keep dividing indefinitely

HOW CANCER CELLS DIFFER FROM NORMAL CELLS

Normal Cells	Cancer Cells
Communicate via chemical signals that influence cell division and program cell death	Lack effective communication
Stop dividing at a certain density (density-dependent inhibition)	Keep dividing regardless of density
Only divide a certain number of times	Are not programmed to die (can be immortal)
Must be anchored to divide	Do not need to be anchored

TUMOR VS. CANCER — NOT INTERCHANGEABLE

- **Tumor (neoplasm)** = an abnormal growth of tissue capable of forming a mass
- **Benign tumor** = does NOT invade surrounding tissue and does NOT spread — not cancer
- **Malignant tumor** = CAN invade surrounding tissue and spread to other parts of the body — IS cancer

How Cancer Makes You Sick

HEALTH CONSEQUENCES

- **Tumor growth** — tumor grows and impairs organ function
- **Metastasis** = cancer cells enter the bloodstream, producing secondary tumors in distant organs. This is what makes cancer deadly.
- **Angiogenesis** = blood vessels are routed to the tumor, depriving normal cells of oxygen and nutrients

Two Types of Cancer-Causing Gene Mutations

Gene Type	Normal Function	When Mutated
Proto-oncogene → Oncogene	Promotes cell division when needed (like a gas pedal)	Stuck in the "ON" position → cell divides non-stop even without a signal
Tumor Suppressor Gene	Stops cell division when needed (like brakes)	Brakes are broken → cell cannot stop dividing

SPECIFIC TUMOR SUPPRESSOR PROTEINS — HIGH YIELD

Protein	Normal Function	Cancer Connection
p53	Inspects DNA integrity; stops cell division if DNA is damaged; triggers DNA repair or apoptosis	**50% of ALL cancers** have a disabled p53. 70–80% of lung cancers have mutant p53.
Rb (Retinoblastoma protein)	Blocks E2F protein, which prevents DNA copying. When Rb is released, E2F enables DNA replication.	**40% of ALL cancers** have a defective Rb. Without Rb, E2F is always active → constant DNA replication.
Telomerase inhibitor	Inhibits telomerase → chromosomes lose telomere tips with each division → limits cell to ~30 divisions	Almost ALL cancers destroy the telomerase inhibitor → cancer cells divide indefinitely

KEY VOCABULARY

- **Metastasis** = cancer cells break away from the original tumor, enter the bloodstream or lymph system, and form new tumors elsewhere
- **Angiogenesis** = tumors send signals to grow new blood vessels toward themselves
- **Apoptosis** = programmed cell death — a normal process; cancer cells often disable it
- **Carcinogen** = any substance or agent that causes cancer (e.g., cigarette smoke, UV radiation)

Cancer Staging — Stages 1–4

EXAM QUESTION — GIVEN A SCENARIO, IDENTIFY THE STAGE

Often staged based on tumor size, number of lymph nodes involved, and the number of distant metastases. Ranges from 1–4. The lower the number, the better the prognosis.

Stage	Where Is the Cancer?	Prognosis
Stage 1	Small tumor, confined to original site — has NOT spread to lymph nodes or other organs	Best — most curable

Stage 2	Larger tumor, still mostly local, may have spread to nearby lymph nodes	Good with treatment
Stage 3	Tumor has spread to more lymph nodes and possibly nearby tissues or organs	More serious
Stage 4	Cancer has metastasized — spread to distant organs (liver, lungs, brain, bones)	Most serious; hardest to treat

DIAGNOSIS METHODS

- CT scan
- MRI

Cancer Treatments

Treatment	What It Does	Key Facts
1. Surgery	Physically removes the tumor	First modality used successfully in cancer treatment. Only curative therapy for many common solid tumors. Works best when there are no distant metastases and no local infiltration.
2. Radiation	Ionizing radiation (X-rays or gamma rays) damages cancer cell DNA	Success depends on sensitivity difference between tumor and normal tissue. Side effects: fatigue, skin reactions, GI toxicity. Radiation is itself mutagenic, carcinogenic, and teratogenic — can cause secondary cancers.
3. Chemotherapy	Drugs that halt cell division throughout the body	Types include alkylating agents, antimetabolites, antitumor antibiotics, plant alkaloids, hormonal agents, and immunotherapy. Every chemotherapeutic has some effect on normal tissue. Kills fast-dividing cells — including hair follicles and gut lining → hair loss, nausea, fatigue.

Quick Reference — Bring This to the Exam

Equations · ATP Counts · Locations · Stages · Key Terms

Master Equations

PHOTOSYNTHESIS

$$6CO_2 + 6H_2O + \text{Light} \rightarrow C_6H_{12}O_6 + 6O_2$$

Endergonic · Thylakoid membrane (light rxns) + Stroma (Calvin cycle)

CELLULAR RESPIRATION

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 36\ \text{ATP}$$

Exergonic · Cytoplasm (glycolysis) + Mitochondria (Krebs + ETC)

Cellular Respiration — ATP Cheat Sheet

Stage	Location	ATP Made	Also Makes	O_2 Needed?
Glycolysis	Cytoplasm	**2 ATP**	2 NADH, 2 Pyruvate	No
Transition (Pyruvate → Acetyl-CoA)	Mitochondrial matrix	0 ATP	2 NADH, 2 CO_2	Yes
Krebs Cycle	Mitochondrial matrix	**2 ATP**	6 NADH, 2 $FADH_2$, 4 CO_2	Yes
Electron Transport Chain	Inner mitochondrial membrane	**34 ATP**	H_2O (water)	Yes — final acceptor
TOTAL (Aerobic)	—	**36 ATP**	—	Yes

Fermentation	Cytoplasm only	**2 ATP**	Ethanol + CO_2 OR Lactic acid	No

Photosynthesis — Location Cheat Sheet

Stage	Location	Reactants	Products
Light Reactions	Thylakoid membrane	H_2O + Light + ADP + $NADP^+$	ATP + NADPH + O_2
Calvin Cycle	Stroma	CO_2 + ATP + NADPH	G3P → Glucose + ADP + $NADP^+$

Mitosis — PMAT Cheat Sheet

Phase	Key Word	What Happens
Prophase	PREPARE	Chromosomes condense; nuclear envelope breaks down; spindle forms
Metaphase	MIDDLE	Chromosomes line up at center (metaphase plate); best time to count chromosomes
Anaphase	APART	Sister chromatids pulled to opposite poles; cell elongates
Telophase + Cytokinesis	TWO	Two nuclei form; cell splits into 2 identical diploid daughter cells

Cancer — Quick Reference

Concept	Key Fact
Oncogene	Mutated proto-oncogene → stuck "gas pedal" → constant cell division
Tumor suppressor	Broken "brakes" → cell cannot stop dividing

p53	DNA damage inspector; disabled in 50% of all cancers; 70–80% of lung cancers
Rb	Blocks DNA copying; disabled in 40% of all cancers
Benign	Does NOT spread — not cancer
Malignant	CAN spread — is cancer
Metastasis	Cancer spreads via bloodstream to distant organs = Stage 4
Angiogenesis	Tumor grows its own blood supply
Stage 1	Local only, small tumor
Stage 2	Nearby lymph nodes
Stage 3	Regional spread, more lymph nodes
Stage 4	Distant metastasis (liver, lungs, brain, bones)
Surgery	Removes tumor — best for Stage 1–2 solid tumors
Radiation	Ionizing radiation damages cancer cell DNA
Chemotherapy	Halts cell division — kills fast-dividing cells (cancer + hair + gut)

Memory Tricks — Optional

Use these if they help — skip them if you don't need them

CHAPTER 6 — WHERE PHOTOSYNTHESIS HAPPENS

"THYLAKOID = LIGHT · STROMA = SUGAR"

Light reactions happen in the thylakoid membrane. The Calvin cycle (sugar-making) happens in the stroma. Light hits the thylakoid → energy goes to stroma → glucose is made.

CHAPTER 6 — CALVIN CYCLE STEPS

"CATCH · REDUCE · RECYCLE"

1. CATCH the CO_2 (carbon fixation by RuBisCO) 2. REDUCE it to G3P using ATP + NADPH 3. RECYCLE the RuBP so the cycle keeps going

CHAPTER 8A — CELL CYCLE ORDER

"I Pee on the MAT"

Interphase → **P**rophase → **M**etaphase → **A**naphase → **T**elophase → Cytokinesis

CHAPTER 8A — SISTER CHROMATIDS VS. HOMOLOGUES

"Sisters = Identical Twins · Homologues = Fraternal Twins"

Sister chromatids = identical twins (exact same DNA, joined at the centromere). Homologous chromosomes = fraternal twins (same chromosome type, but from different parents — similar but not identical).

CHAPTER 8B — P53 AND RB

"p53 = Police · Rb = Red light"

p53 is the DNA police — checks for damage and arrests the cell cycle. Rb is a red light — stops the cell from copying DNA. When either is broken, the cell runs red lights and ignores the police → cancer.

CHAPTER 8B — THE 3 CANCER TREATMENTS

"CUT · BURN · POISON"

Surgery = CUT out the tumor. Radiation = BURN the cancer cells with ionizing energy. Chemotherapy = POISON fast-dividing cells.